

3)Write a Java program to demonstrate the handling of ArithmeticException, ArrayIndexOutOfBoundsException, and NumberFormatException using appropriate catch blocks.

PROGRAM:

```
public class Main {

    public static void main(String[] args) {

        // ArithmeticException
        try {
            int a = 10 / 0;
        } catch (ArithmeticException e) {
            System.out.println("ArithmeticException: Cannot divide by zero");
        }

        // ArrayIndexOutOfBoundsException
        try {
            int arr[] = {10, 20, 30};
            System.out.println(arr[5]);
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("ArrayIndexOutOfBoundsException: Invalid array index");
        }

        // NumberFormatException
        try {
            String str = "ABC";
            int num = Integer.parseInt(str);
        } catch (NumberFormatException e) {
            System.out.println("NumberFormatException: Invalid number format");
        }
    }
}
```

```
    }  
  }  
}
```

OUTPUT:

ArithmeticException: Cannot divide by zero

ArrayIndexOutOfBoundsException: Invalid array index

NumberFormatException: Invalid number format
